LIRUI WANG

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EDUCATION

University of Washington, Seattle, WA *Sep 2016 - Present*
Bachelor of Science, Electrical Engineering, (Honors), College of Engineering
Bachelor of Science, Computer Science, (Honors), College of Arts and Sciences
Bachelor of Science, Mathematics, (Minor), College of Arts and Sciences
GPA: 3.97/4.00, Advisor: Prof. Dieter Fox

RESEARCH PROJECTS

Joint Motion and Grasp Planning *March 2019 - Present*

- Implement a network that combines data-driven and traditional method to synthesize grasp pose.
- Prototype RL, IL, and MPC with PyBullet environment to map observed points directly to control in task space.
- Implement CHOMP Planner and introduce an integrate planner that optimizes manipulation trajectory with online grasp synthesis and selection.
- Propose an online learning framework and a grasp synthesis approach to generate grasp and select grasp online.

Pose Tracking with the Deep Iterative Matching Network *June 2018 - March 2019*

- Tested DeepIM on YCB datasets and achieved robust results trained on synthetic data only.
- Improved network training stability, pose distribution and occlusion generation.
- Extended DeepIM in real-time tracking for occluded scene in YCB and built AR demos.
- Built renderer and kinematics tree for vision system application on robotics.

Unsupervised Flow and Depth Methods Study *March 2018 - June 2018*

- Investigated a few deep learning methods for 3D geometry tasks such as depths, flow, and egomotion.
- Explored unsupervised structures and ideas such as SfMlearner and MonoDepth and analyzed them on a robotic setting.

Baxter Hand-Eye Coordination *Aug 2017 - March 2018*

- Worked on a manipulation task with PoseCNN, which enables Baxter to move daily objects.
- Studied on camera models, hand-eye calibration with depth camera, robot kinematics, 3D geometry and ROS packages for tracking and control.

WORK EXPERIENCE

Seattle, WA *Sep 2019 - Present*
Computer Vision Teaching Assistant, UW CSE 455

Shenzhen, China *Jun 2017 - Aug 2017*
Robotics Engineer Intern, DJI Technology

- Worked on autonomous robot battle which uses vehicles with TX1 onboard and DJI Manifold drones with Guidance
- Implemented functions such as take-off, tag detection, and refilling to localize drone and communicate

- Applied SLAM to mapping and localize, and built a top-level state machine to strategize and navigate the robot

LEADERSHIP AND INVOLVEMENT

Seattle, WA Sep 2016 - Feb 2018
Computer Vision Lead *Advanced Robotics at the University of Washington*

- Managed a team of 10 students to design computer vision tools for our robots in a competition RoboMasters hosted by DJI.
- Performed system integration of controls, navigation and serial communication

MISCELLANEOUS PROJECTS

- **Sound Scape**: Designed a sound processing pipeline from scratch for modelling the acoustics of a train whistle
- **Music Editor**: Created a demo app that can record sound pieces and translate into sheet music display UI using DTFT, and it was awarded 1st place out of 35 teams in Seattle
- **Drowsiness Detection**: Built an embedded system on Raspberry Pi that that simulates a real-time drowsiness detection system based on human face detection.
- **Robust Deep Learning**: Investigate the method of meta-learning to reweight examples on a fossil dataset, compare it with randomized smoothing technique, and improve the performance with adversarial training.
- **Machine Learning in Robot Trajectory Optimization**: Summarize the formulation of trajectory optimization as functional gradient descent using Reproducing Kernel Hilbert Space (RKHS) and as probabilistic inference using Gaussian Process (GP).

HONORS

- Annual Deans List, University of Washington, *2016 - 2020*
- Tau Beta Pi Scholarship *2017 - 2019*
- Lawrence & Lucille Frey Endowed Electrical Engineering Scholarship *2017 - 2018*
- Patricia G. Lynch and Theodora & Eugene Russell Memorial Computer Science Scholarship *2018 - 2019*

PUBLICATIONS

- **Manipulation Trajectory Optimization with Online Grasp Synthesis and Selection**
  Lirui Wang, Yu Xiang, Dieter Fox
  In Submission of The International Conference on Automated Planning and Scheduling (ICAPS 2020)

COURSES

- **Computer Science**: Data Structure and Parallelism, Algorithms, Computer Vision, Artificial Intelligence, Deep Learning, Modern Algorithm Toolbox, Graphics, Robotics, Robustness in Machine Learning (Grad)
- **Electrical Engineering**: Digital Signal Processing, Control Theory, Control Capstones, Embedded System
- **Math**: Multivariable Calculus, Probability and Statistics, Numerical Analysis, Linear Analysis, Topology, Convex Optimization and Analysis (Grad), High Dimensional Statistics and Statistical Learning (Grad), Statistical Learning: Modeling, Prediction, And Computing (Grad)

- **English**: Technical Writing, Advanced Technical Writing in Electrical Engineering

SKILLS

- **Systems**: ROS, Ubuntu, ARM, Arduino, Nvidia TX Series, STM32
- **Languages**: Python, C++/C, Matlab, Java
- **Tools**: Git, CMake, OpenCV, DJI Guidance, Bash, Latex, Tensorflow, Pytorch, MXNet, OpenGL, CUDA, Qt, Bullet, VTK,
- **Hardware**: 3D Printing, Laser Cutting, Machining, Soldering

EXTRACURRICULARS

- **Mens Pickup Soccer Team**: Played regularly at pickup soccer and joined the intramural soccer competition.
- **Math Club**: Joined the meetings to solve problems with other students in the group.